Req ID	Type	Requirement	Acceptance Criterion	Justification (Link to Standard)
R-TRACK-01	FR	Robot shall maintain lateral tracking error within tolerance during S1 nominal scenario.	$\lvert e_{line} \rvert < 0.15$ m for all $t \in [0,75]$ s	Per ISO 12100 Cl. 5.2: Predictable motion requirement for safe operation. Per ISO 9241-210 Cl. 5.2: Understanding operational context defines this core functional need.
R-TRACK-02	FR	Robot shall minimize cumulative tracking error over the full S1 scenario.	IAE ≤ 0.025 m·s	Per ISO 9241-210 Cl. 5.3: Translating user need for precision into measurable target. Per ISO/IEC 25010 Cl. 7.2.1: Performance efficiency requirement for tracking accuracy.
R-OBS-01	FR	Robot shall avoid collision with static obstacles during S2 scenario.	Number of obstacle contacts = 0	Per ISO 12100 Cl. 5.4: Risk reduction through protective measures. Per ISO 9241-210 Cl. 5.2: Context includes unexpected obstacles, so system must handle them.
R-OBS-02	FR	Robot shall complete the S2 scenario after obstacle avoidance manoeuvre.	Scenario completed without manual intervention	Per ISO 9241-210 Cl. 5.6: Design must be evaluated on ability to handle complete scenarios. Per ISO/IEC 25010 Cl. 7.3.1: Recoverability requirement for system continuity.
R-SAFE-01	FR	Robot shall limit speed during sensor fault condition.	$v \leq 0.45$ m/s when fault_flag = 1	Per IEC 61508 Cl. 7.4.2.3: Safe state requirement for fault conditions. Per ISO 9241-210 Cl. 5.6: Safety is key usability outcome - system must behave predictably in fault conditions.
R-PERF-01	NFR	Robot shall complete S1 scenario within specified time limit.	$t_{final} \leq 75$ s	Per ISO 9241-210 Cl. 5.3: User efficiency requirement. Per ISO/IEC 25010 Cl. 7.2.5: Time behaviour performance requirement.
R-ENERGY-01	NFR	Controller shall minimize control energy usage.	Energy proxy ≤ 60	Per ISO 14040 Cl. 4.3.1: Life cycle impact assessment - energy consumption is key environmental indicator. Per ISO 9241-210 Cl. 5.2: Part of broader context includes environmental impact.
R-ROBUST-01	NFR	Controller shall detect and respond to temporary sensor dropout.	Fault detected within 1 control cycle	Per ISO 9241-210 Cl. 5.6: Reliability impacts user trust. Per ISO/IEC 25010 Cl. 7.2.2: Fault tolerance requirement - system that fails silently is not user-centered.
R-HMI-01	NFR	HMI shall display real-time telemetry (e_{line}, v, ω, flags) during simulation.	All variables update at ≥10 Hz, clearly labeled	Per ISO 9241-210 Cl. 5.3: Direct translation of user need for "clear feedback" into measurable requirement. Per ISO 9241-210 Cl. 5.2: Understanding user's need to monitor system state.

Standard	Clause	Application
ISO 9241-210:2019	Cl. 5.2, 5.3, 5.6	Human-centred design (Primary CA1 Standard)
ISO 12100	Cl. 5.2, 5.4	Safety of machinery
IEC 61508	Cl. 7.4.2.3	Functional safety
ISO 14040	Cl. 4.3.1	Life cycle assessment
ISO/IEC 25010	Cl. 7.2.1, 7.2.2, 7.2.5, 7.3.1	System and software quality requirements